Applied Geodata Science 2

Digital soil mapping

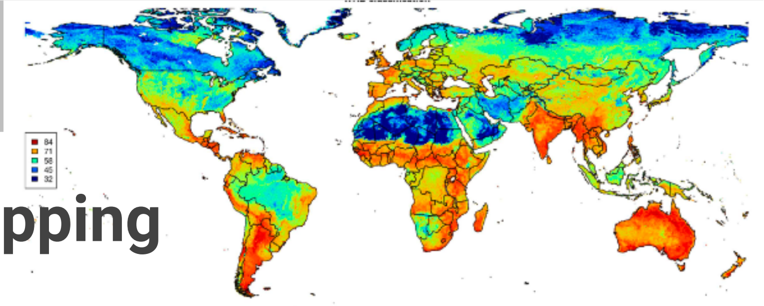
Prof. Dr. Benjamin Stocker

Fall semester 2025

GECO

UNIVERSITÄT BERN

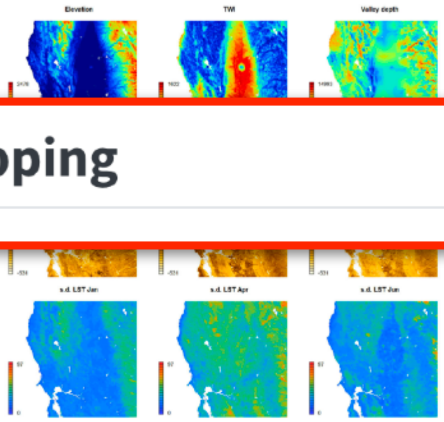
A: Making sense of information



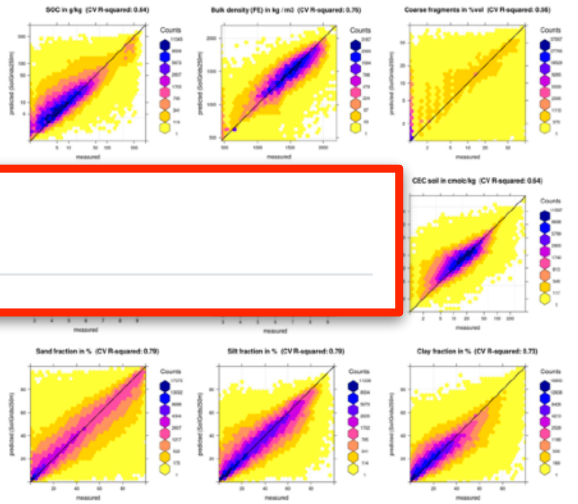
- **Spatial upscaling for digital soil mapping**
  - **Train: pixel-wise:** Y = *f*(**x**)
  - **Predict: pixel-wise:** Y = *f*(**x**’)

Block 3: Digital soil mapping

Training data:
150,000 soil profiles



Environmental covariates
with global coverage



Out-of-sample prediction
and globally upscaled
product

Hengl et al., 2017

Soil forming factors

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Soil forming factors

$$S = f(C, O, R, P, T)$$

- S = soil
- C = climate → Mean climatology data (e.g., WorldClim)
- O = organisms → Remote sensing data (e.g., NDVI)
- R = relief (topography) → Digital elevation model
- P = parent material → Bedrock geology map
- T = time → Time of last glaciation

‘Soil’ is a set of measurable properties

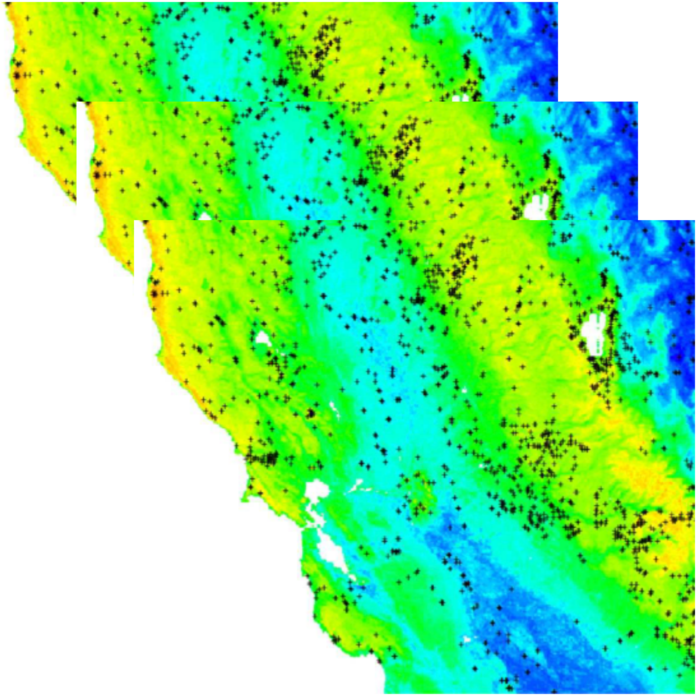
- Organic carbon content
- Bulk density
- Cation exchange capacity
- pH
- Soil texture fractions (silt, clay, sand)
- Coarse fragments

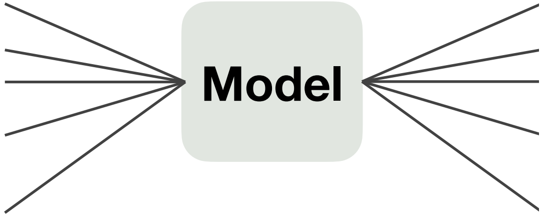
Model

- *C* = climate
- *O* = organisms
- *R* = relief (topography)
- *P* = parent material
- *T* = time

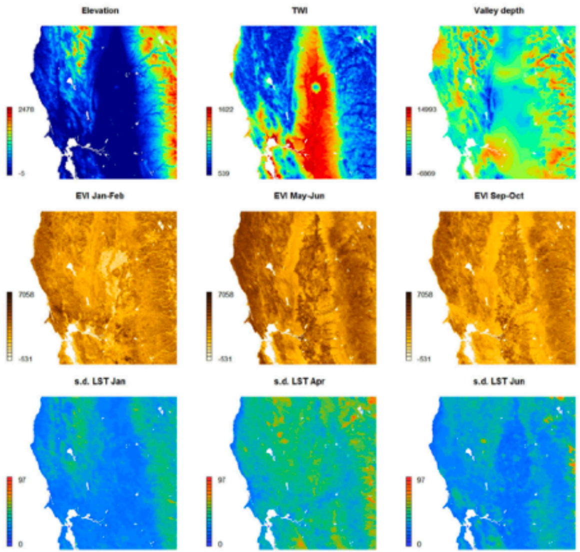
Spatial coverage in predictors - spatial coverage in target variables

Target variables



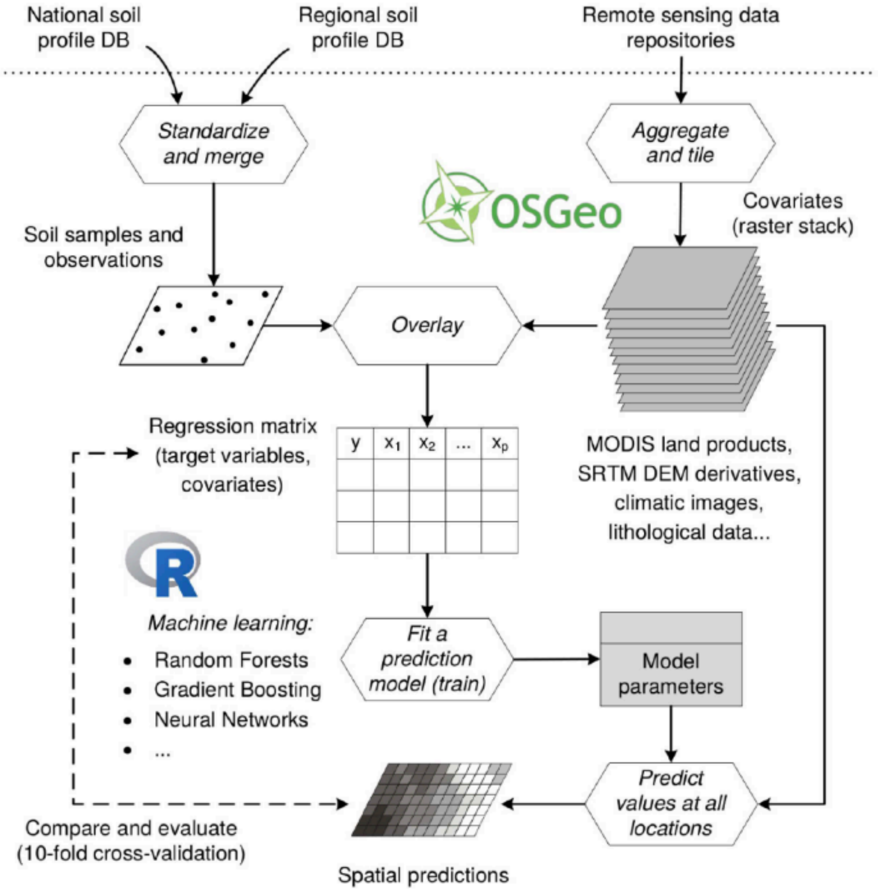


Predictors



Hengl et al., 2017 *PLoS ONE*

Workflow

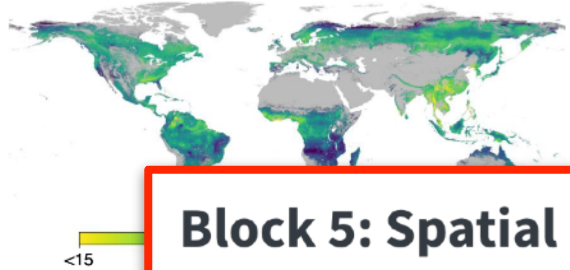


Hengl et al., 2017 *PLoS ONE*

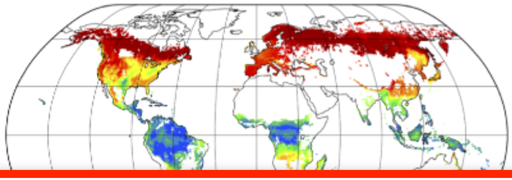
A: Making sense of information

- **Spatial upscaling everything**

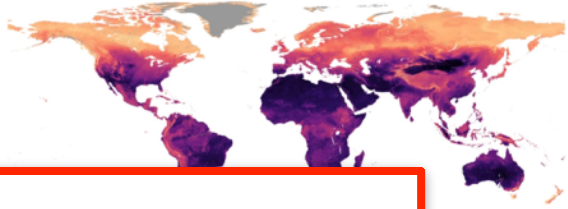
Root mass fraction (%)



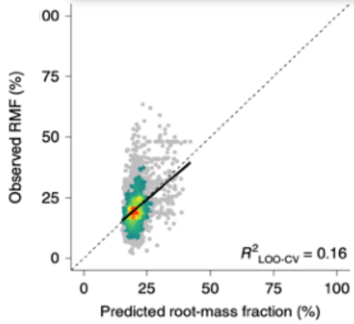
Fraction of trees associated with ectomycorrhizal fungi (%)

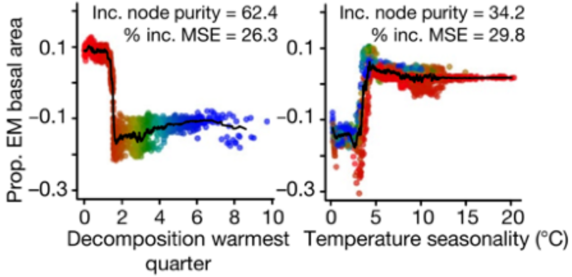


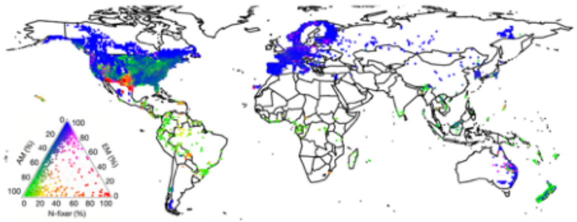
Soil nematode abundance



Block 5: Spatial upscaling





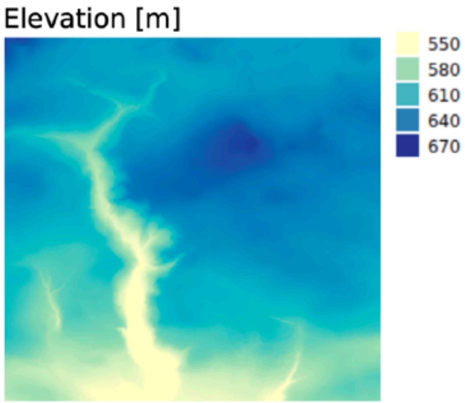


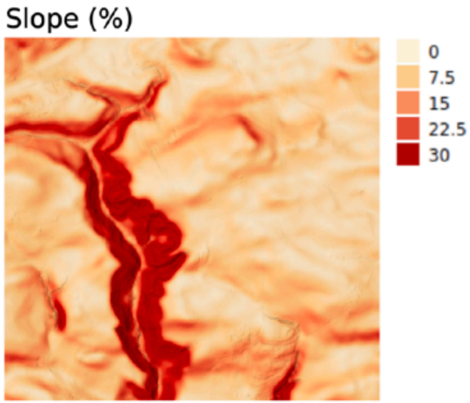
Ma et al., 2020 *Nature Eco. Evo.*

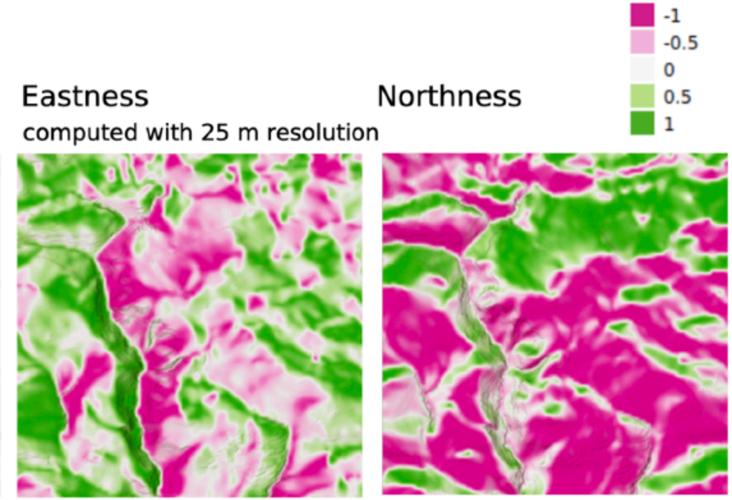
Steidinger et al., 2019 *Nature*

Predictors

Digital elevation model

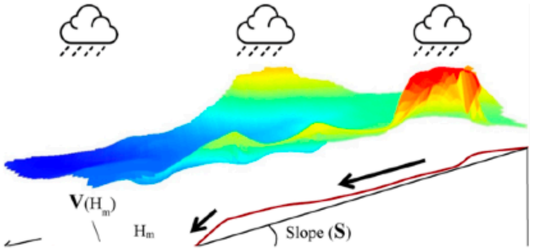






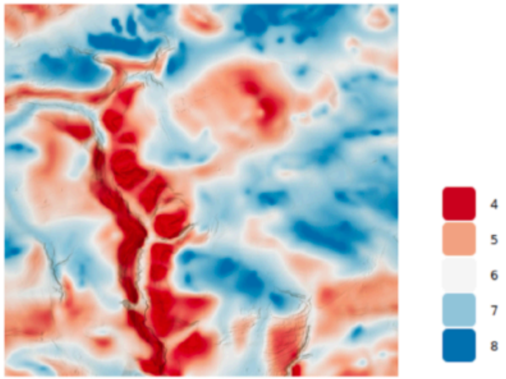
Predictors

Terrain wetness index



$$\mathrm{TWI} = \ln\left(\frac{a}{\tan(\beta)}\right)$$

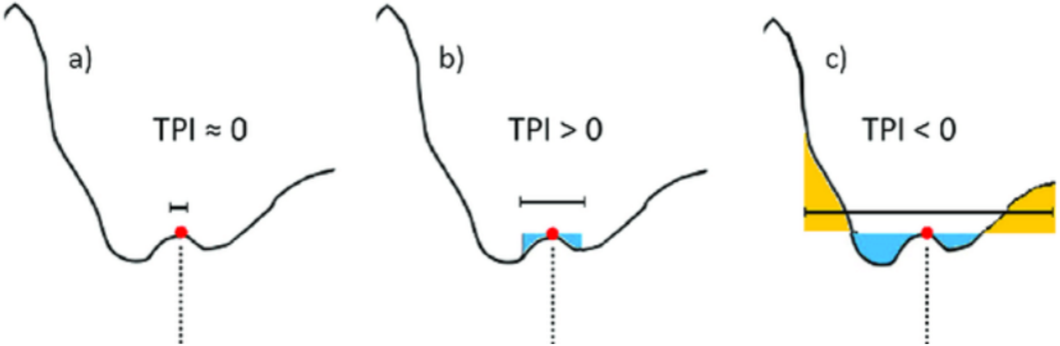
TWI



- **High TWI**: tendency for subsurface moisture convergence, wet soils, shallow groundwater. Valley bottoms.
- **Low TWI**: tendency for subsurface moisture divergence, dry soils, deep groundwater. Ridge positions.

Predictors

Topographic position index



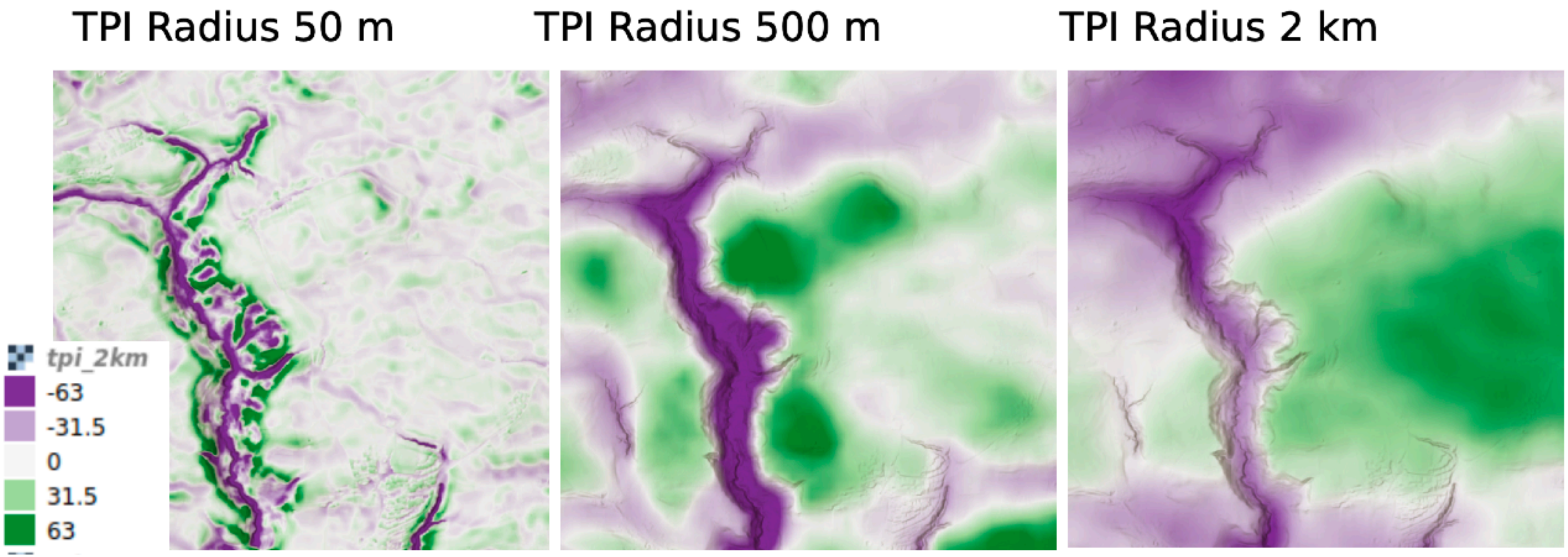
Small scale

Large scale

Salinas-Melgoza et al., 2018 *Ecosphere*

Predictors

Topographic position index



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RESEARCH ARTICLE

SoilGrids250m: Global gridded soil information based on machine learning

Tomislav Hengl[1]*, Jorge Mendes de Jesus[1], Gerard B. M. Heuvelink[1], Maria Ruiperez Gonzalez[1], Milan Kilibarda[2], Aleksandar Blagotić[3], Wei Shangguan[4], Marvin N. Wright[5], Xiaoyuan Geng[6], Bernhard Bauer-Marschallinger[7], Mario Antonio Guevara[8], Rodrigo Vargas[8], Robert A. MacMillan[9], Niels H. Batjes[1], Johan G. B. Leenaars[1], Eloi Ribeiro[1], Ichsani Wheeler[10], Stephan Mantel[1], Bas Kempen[1]

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SOIL, 4, 1–22, 2018
https://doi.org/10.5194/soil-4-1-2018

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Evaluation of digital soil mapping approaches with large sets of environmental covariates

Madlene Nussbaum[1], Kay Spiess[1], Andri Baltensweiler[2], Urs Grob[3], Armin Keller[3], Lucie Greiner[3], Michael E. Schaepman[4], and Andreas Papritz[1]

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Machine learning based soil maps for a wide range of soil properties for the forested area of Switzerland

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